Unit 6, Lesson 2: Prime Factorization

Benchmark

MA.6.NSO.3.4 Express composite whole numbers as a product of prime factors with natural number exponents.

Learning Target

- [ ] I can express composite whole numbers as a product of prime factors.

6.2.1 Warm-Up: Would You Rather?

1. Which of the following would you rather choose? Justify your reasoning using mathematics.

Deposit $\$3$ in the bank and have it triple each week for 4 weeks.	Deposit $\$4$ in the bank and have it quadruple each week for 3 weeks.

6.2.2 Collaboration: Exploring Factors

Work with your partner on the following.

1. Write the number 24 as a factor pair that uses the greatest factor other than itself.

2. Write the number 12 as a factor pair that uses the greatest factor other than itself.

3. Write the number 6 as a factor pair that uses the greatest factor other than itself.

4. Write the number 3 as a factor pair that uses the greatest factor other than itself.

5. How did each of the above tasks, problems 2-4, relate to the factor pair that came before it?

6. What was different about rewriting the number 3 as a factor pair using the greatest factor other than itself?

6.2.3 Guided Instruction: Prime Factorization

Each of the factor pairs from the Collaboration can be used to rewrite 24 using more than one pair of factors.

For example, the first factor pair was $2 \cdot 12$, so 24 can be rewritten as $24 = 2 \cdot 12$.

Because 12 can be written as $2 \cdot 6$, 24 can also be rewritten as $24 = 2 \cdot 2 \cdot 6$.

1. Since 6 can be expressed as __________, then 24 can be rewritten as $24 =$ ________________________.

This is called the ***prime factorization*** of 24 because all of the factors are ***prime numbers***.

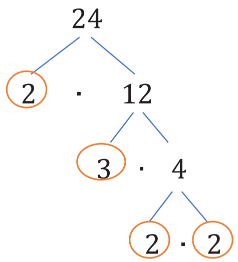
A ***prime number*** is a whole number greater than 1 that is not divisible by any whole number other than 1 and itself.

A ***prime factor*** is a factor that is a prime number.

Prime factorization is the expression of a number as the product of prime factors.

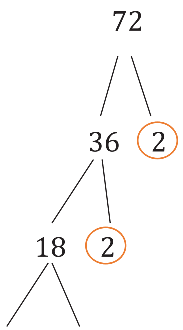
2. Cory determined the prime factorization of 24 using different factor pairs.

   a. Explain whether Cory's method changed the final result.



   b. Why did Cory circle the values he did?

3. Find the prime factorization of 72 by completing the work below.



   The prime factorization of 72 is ______________________.

4. Consider the expression $2 \times 2 \times 2 \times 3 \times 3 \times 5$.

   a. What is the product of $2 \times 2 \times 2 \times 3 \times 3 \times 5$?

   b. The prime factorization of ________ is $2 \times 2 \times 2 \times 3 \times 3 \times 5$.

Numbers with multiple repeating prime factors can be written using exponents. For example, the prime factorization of 360 can be written as $2^3 \times 3^2 \times 5$.

5. Why is it helpful to write the prime factorization with exponents?

6.2.4 Guided Practice: Prime Factorization

1. Find the prime factorization of each value. Use exponents where appropriate.

   a. 132 b. 270 c. 882

2. Sophie said the prime factorization of 210 is $3 \cdot 7 \cdot 10$.

   a. Determine the error Sophie made. Then, write her a brief note explaining her error and how she can learn from it.

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   I
   Start
   To
   Acquire
   Knowledge
   Experience
   Skills

   b. What is the prime factorization of 210?

6.2.5 Your Turn!

1. Find the prime factorization of each value. Use exponents where appropriate.

   a. 904 b. $1{,}125$ c. 80

2. Which number includes the largest exponent when written in prime factorization? Explain your answer.

   $\boxed{5{,}670}$ $\boxed{64}$

3. Find a number that meets the following conditions.

   - The value of the number is less than 150.
   - One of its prime factors has an exponent of 3.
   - When expressed in exponential form, its prime factorization has only two unique bases.

6.2.6 Wrap-Up: Ticket Out the Door

1. Find the prime factorization of each value. Use exponents where appropriate.

   a. 168

   b. $1{,}040$

   c. 144

Unit 6, Lesson 3: Adding Integers Using Integer Chips

Benchmark

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

Learning Targets

- ☐ I can express integer addition using integer chips.
- ☐ I can add integers using integer chips.